

Basic Civil and Mechanical Engineering

SITAMS CSE (AI & ML) • R23 • I Year • I Semester • Course Code: 23ESC111T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit Part A • Unit 1: Introduction to Civil Engineering

Syllabus topics

- Basics of civil engineering
- overview of civil engineering
- materials and building construction fundamentals

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part A • Unit 2: Surveying and Foundation Engineering

Syllabus topics

- Surveying objectives and basic methods
- foundation engineering fundamentals

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part A • Unit 3: Transportation, Water Resources and Environmental Engineering

Syllabus topics

- Basic transportation systems
- water resources
- environmental engineering fundamentals

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part B • Unit 1: Introduction to Materials and Manufacturing Engineering

Syllabus topics

- Role and scope of mechanical engineering
- ferrous and non-ferrous metals, ceramics, composites and smart materials
- casting, forming, joining, machining, CNC, 3D printing and smart manufacturing

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part B • Unit 2: Introduction to Thermal Engineering

Syllabus topics

- Boilers
- refrigeration and air-conditioning cycles
- refrigerants
- vapour-compression and absorption systems
- air-conditioning systems
- Otto and Diesel cycles
- IC engines
- electric and hybrid vehicles

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part B • Unit 3: Power Plants, Mechanical Transmission and Robotics

Syllabus topics

- Steam, diesel, hydro, gas-turbine and nuclear power plants
- belt, chain, rope and gear drives
- robotics joints, links, configurations and applications

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.